Reg. Number:	22BAI1266

VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Continuous Assessment Test (CAT) – I AUGUST 2025

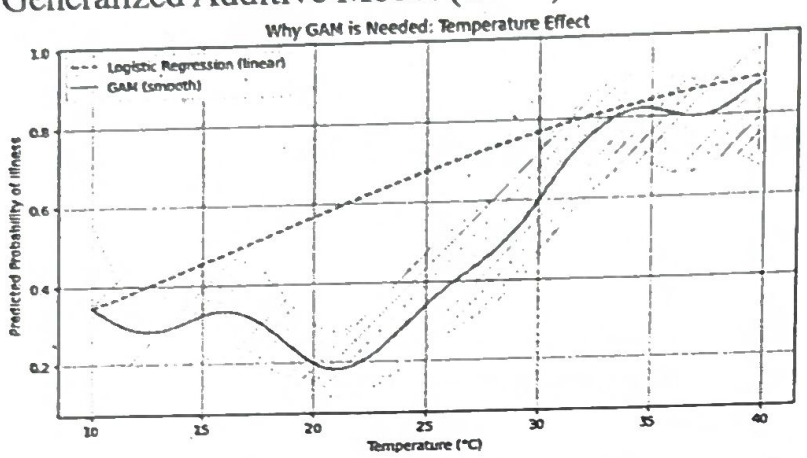
Programme	:	B.Tech CSE	Semester	:	Fall Sem 2025-2026
Course Code & Course Title	:	BCSE418L - Explainable Artificial Intelligence	Class Number	:	CH2025260102065 CH2025260102067 CH2025260102523
Faculty	:	Dr. VINOTHINI A Dr. POONKODI Dr. POORNIMA S	Slot	:	E1
Duration	:	90 mins	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1		Explain the Explainable AI (XAI) methods with respect to Fidelity, Understandability, Sufficiency, Low Overload, and Efficiency. [5 marks] Which of the above criteria have been violated in the following scenario. Justify your classification.[5 marks] a) Ramesh has provided explanations for his financial model in English. But his customers do not know English. b) Rekha has generated explanations for the Deep Learning Model. But the explanations are generated from an interpretable model built independently on the same training data.	10	1	2
2		Regulating traffic to reduce congestion is one of the major concerns in building a smart city. To formulate traffic, the system collects data from traffic cameras, GPS sensors in public transports, weather sensors, and reports on road accidents and presents the following patterns. • Heavy rainfall and slower traffic often coincides. • Construction of bridges or roads increases travel times. • Due to local festivals in certain regions spikes in congestion occur. Even without the above said reasons congestion may occur due to an increase in the number of vehicles at peak hours. i. Identify all the possible causal links. (3 marks) ii. How to apply causality learning techniques to learn these links from data. (4 marks)	10	1	3

		iii. Discuss how the learned model could be used to run intervention simulations. (3 marks)			
3		A public health researcher is studying heat-related illness among outdoor workers, aiming to determine whether workers developed heat-related illness (Yes/No). The dataset includes Temperature (°C) as the independent variable and illness status as the dependent variable. Initially, the researcher fits a simple logistic regression model, assuming a linear relationship between temperature and the log-odds of illness. However, this model underestimates the risk of illness. The researcher then fits a Generalized Additive Model (GAM).  i) Why did the researcher find GAM to be more appropriate than a simple logistic regression? (3 marks) ii) How could GAM handle the non-linear relationships observed for temperature and illness risk? (4 marks) iii) How does GAM support interpretability? (3 marks)	10	2	3
4		Consider the decision tree given in Fig 1 below. This tree is built to predict a person's heart risk as either: Low Risk or High Risk. It uses 3 input features: Smoker (0 = No, 1 = Yes), Age (in years), Weight (in kg) (i) How do you interpret the feature importance in decision tree? (4 marks) (ii) Explain the predictions from the above decision tree for the given instance. (6 marks) a. Smoker=no, Age=15, Weight=55 b. Smoker=yes , Age=45, Weight=80. c. Smoker=yes, Age=17, Weight=65.	10	2	3
5		You are a part of a team developing an interpretable soil classification model based on soil images to classify soil types as clay, sandy, or loamy. Before deploying the model, your team wishes to evaluate how well the explanations provided by the model help different users understand and trust its predictions.	10	2	3

		i. Describe three different levels of evaluation to evaluate the interpretability of the soil classification model. (5 marks) ii. Assume that the soil classification model is built using Convolution Neural Networks. Discuss the techniques that can be used to interpret the predictions made by the CNN model.(5 marks)			

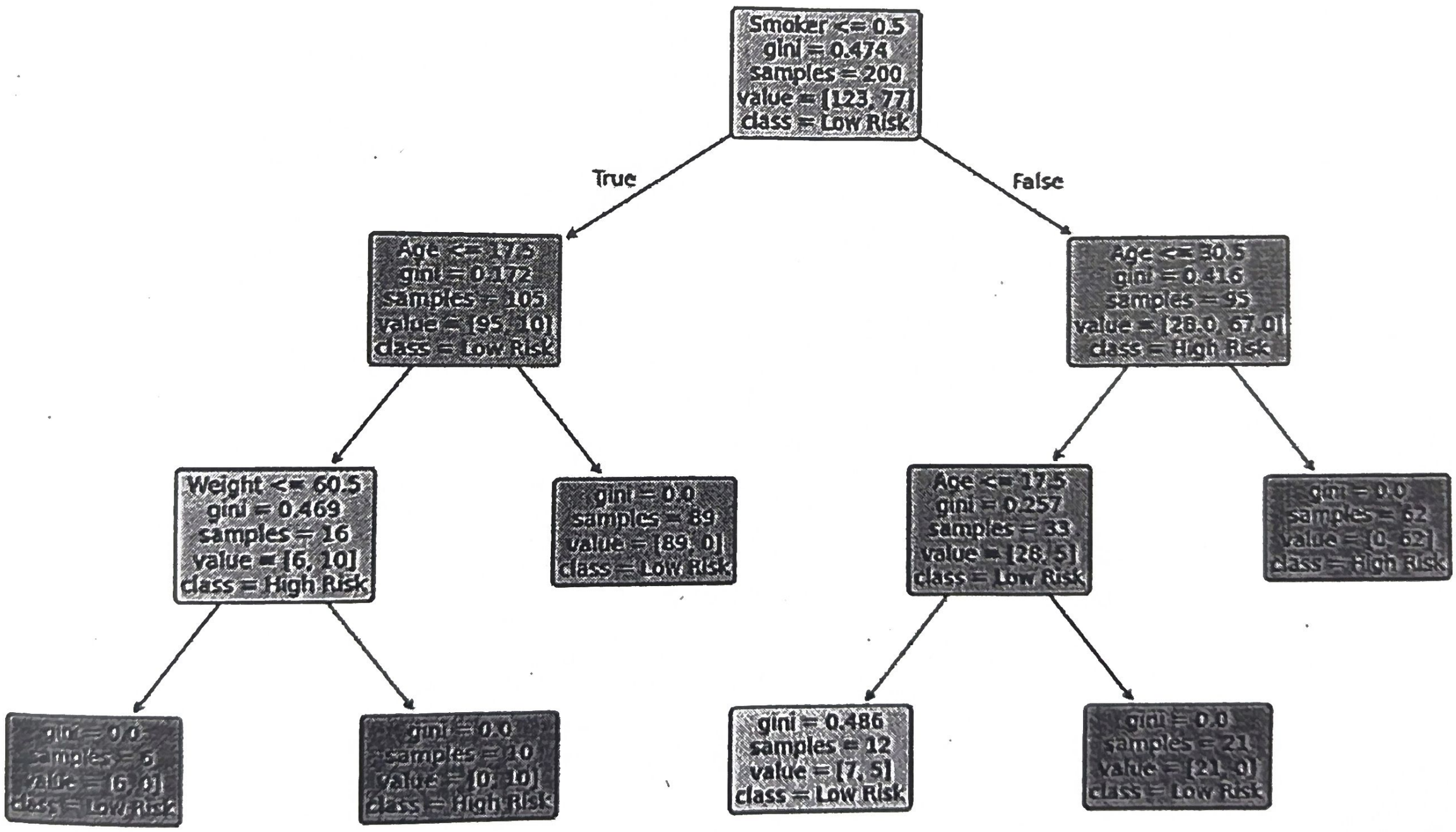


Fig 1. Decision Tree of Q.no 4

**********All the best **********